

Measurement-updated Stokes prediction of E_4 visibility

Hybrid MZI: comparison of isolated-state polarization measurements with a swept-PAX power measurement

1. Minimal update to the ideal model. The ideal path- E_4 field contains two coherent contributions whose relative phase is controlled by ϕ_1 . Replacing the ideal orthogonal basis states by the *measured* normalized polarization states $|A\rangle$ and $|B\rangle$ gives

$$E_4(\phi_1) = \sqrt{P_A} e^{i\phi_1} |A\rangle + \sqrt{P_B} |B\rangle. \quad (1)$$

The corresponding intensity is

$$I_4(\phi_1) = P_A + P_B + 2\sqrt{P_A P_B} \Re[e^{i\phi_1} \langle B|A\rangle], \quad (2)$$

so the theoretical fringe visibility is

$$V_{\text{th}} = \frac{2\sqrt{P_A P_B}}{P_A + P_B} |\langle A|B\rangle|. \quad (3)$$

For fully polarized states, the Jones-state overlap is obtained directly from the normalized Stokes vectors,

$$|\langle A|B\rangle| = \sqrt{\frac{1 + \mathbf{s}_A \cdot \mathbf{s}_B}{2}}. \quad (4)$$

2. Numerical substitution from the realigned isolated-state measurement. The measured states and powers were

$$\begin{aligned} \mathbf{s}_A &= (+0.990466, -0.051196, +0.127887), & P_A &= 0.342\,329 \text{ mW}, \\ \mathbf{s}_B &= (-0.989706, -0.092665, -0.109066), & P_B &= 0.417\,919 \text{ mW}. \end{aligned}$$

Therefore

$$\begin{aligned} \mathbf{s}_A \cdot \mathbf{s}_B &= -0.989475, \\ |\langle A|B\rangle| &= \sqrt{\frac{1 - 0.989475}{2}} = 0.072544, \\ \frac{2\sqrt{P_A P_B}}{P_A + P_B} &= 0.995045, \end{aligned}$$

which gives

$$V_{\text{th}} = (0.995045)(0.072544) = 0.07219 \approx \mathbf{7.22\%}. \quad (5)$$

Thus the measured power imbalance is a small correction; the dominant predicted contrast comes from the measured nonorthogonality of the two polarization states.

3. Visibility measured directly by the PAX during the ϕ_1 sweep. The swept acquisition recorded $S_0 = P_{\text{PAX}}$ simultaneously with the Stokes trajectory. After excluding the initial ~ 1 s settling region, the observed power span was approximately

$$P_{\text{min}} = 0.7264 \text{ mW}, \quad P_{\text{max}} = 0.8615 \text{ mW},$$

corresponding to a literal-extrema visibility of $V_{\text{raw}} = 8.51\%$. Because the PAX samples only about 12 points per actuator period, cycle-wise extrema tend to miss the true extrema: the repeated-cycle mean is $6.60\% \pm 0.61\%$. A pooled 2–98% power envelope gives $V_{\text{PAX}} = 7.23\%$.

Quantity	Visibility	Interpretation
Stokes/Jones prediction	7.22%	No fit; isolated measured states and powers
PAX pooled 2–98% envelope	7.23%	Robust repeated-sweep estimate
PAX cycle-wise mean	$6.60\% \pm 0.61\%$	Sparse phase sampling lowers extrema
PAX literal global extrema	8.51%	Most sensitive to individual samples

Result. The simplest measurement-updated Jones model predicts $V_{\text{th}} = 7.22\%$. The independently swept PAX power data place the observed contrast in the same $\sim 7\%$ range; the robust pooled estimate is 7.23%. Within the sampling limitations of the PAX sweep, the measured polarization nonorthogonality is therefore quantitatively sufficient to account for the visibility observed by the PAX in E_4 .